Demo Video Voiceover Script

Autonomous Volcanic Terrain Exploration Using a Markov Decision Process

CSE 440, Section 1, Group 5 · Shakil Ahmed, Fahim Foysal, Shefa Tabassum, Tanvir Ahmed

The video (`others/demo_video.mp4`) is exactly 60 seconds. Each block below is timed to what is on screen at that moment. Record each block separately against the video and trim words rather than speeding up if a block runs long. All numbers are the real values the video shows.

Script

0:00 – 0:02 — the command is typed

This is our autonomous volcanic explorer. We run the whole project with one command: python main dot py.

0:02 – 0:20 — pipeline output scrolls

First, the terrain generator builds a random ten-by-ten volcanic map: safe ground, lava, craters, gas zones, rock, science points, and the base station. Then the MDP is built from it, with about eighty-five states and a discount factor of point nine, and value iteration computes an optimal action for every single cell. You can see that policy printed as a grid of arrows. Finally the agent runs its mission and prints a summary: steps taken, reward, hazards, science collected, and whether it survived.

0:20 – 0:34 — mission playback window

Here is that mission replayed step by step. The rover leaves the base and follows the policy, but the world is changing around it: gas clouds drift and lava spreads into safe ground. Every time the terrain changes, the agent re-plans with a fresh round of value iteration. When it reaches a science point, it collects the sample once, marked with a gold star, and re-plans again for the next objective.

0:34 – 0:40 — mission map image

The final mission map: twenty-four steps, two samples collected, seventy-one terrain changes survived, and the rover is still alive.

0:40 – 0:50 — experiments command and results table

Now the comparison. We run the same twenty random terrains with three agents: our MDP agent, a greedy explorer that ignores soft hazards, and a random walker.

0:50 – 0:57 — performance plot

Only the MDP agent earns positive reward, plus forty-two against minus one twenty-four and minus one eighty. It survives sixty percent of missions, twice the greedy agent, while entering five times fewer hazards.

0:57 – 1:00 — closing line

Every run is seeded, so all of this is fully reproducible from the repository.

Timing at a glance

Time	On screen	Words
0:00 – 0:02	Typing `python main.py`	19
0:02 – 0:20	Terrain grid, MDP policy, mission summary	78
0:20 – 0:34	Animated mission playback (dynamic hazards)	70
0:34 – 0:40	Final mission map	20
0:40 – 0:50	Experiments command and summary table	28
0:50 – 0:57	Performance plot	32
0:57 – 1:00	Closing terminal line	13
Total		about 260

How to deliver it

- Pace. About 260 words in 60 seconds is on the fast side. If a block runs over, cut words instead of talking faster. The 0:02 to 0:20 block is the tightest; if needed, drop the sentence beginning "You can see that policy…".
- Tone. Confident and plain, like explaining to a classmate rather than reading aloud. Go slightly slower and lower on numbers so they land.
- Match the screen. Say "gold star" when the first star appears (around 0:26), and say the plus forty-two figure while the reward bars are on screen (0:50 onward). The audio should feel like it is pointing at things.
- Splitting among members. A natural split that matches who built what: Member 1 (Shakil) 0:00 to 0:20, terrain and MDP core; Member 2 (Fahim) 0:20 to 0:34, agent and mission; Member 3 (Shefa) 0:34 to 0:40, the mission map; Member 4 (Tanvir) 0:40 to 1:00, experiments and results.
- Recording. Quiet room, microphone 15 to 20 cm away, half a second of silence at the start and end of each block so cuts are clean. iMovie, DaVinci Resolve, or QuickTime can lay the audio over the video without touching the picture.
- Export. Keep the final audio at or under 60 seconds and re-export as MP4 to the same path, `others/demo_video.mp4`, so the README and report links stay valid.